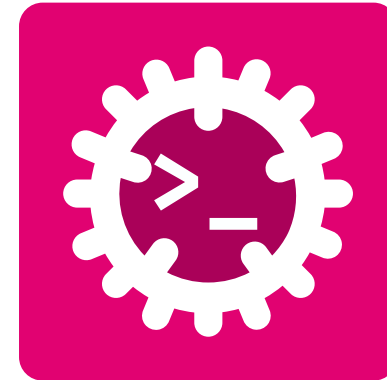


# Advanced Programming

## Concurrency // Parallel Programming



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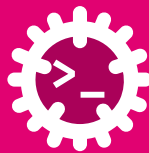


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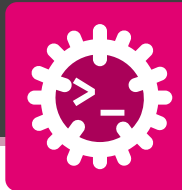


# Concurrency

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Doing more than one thing at once

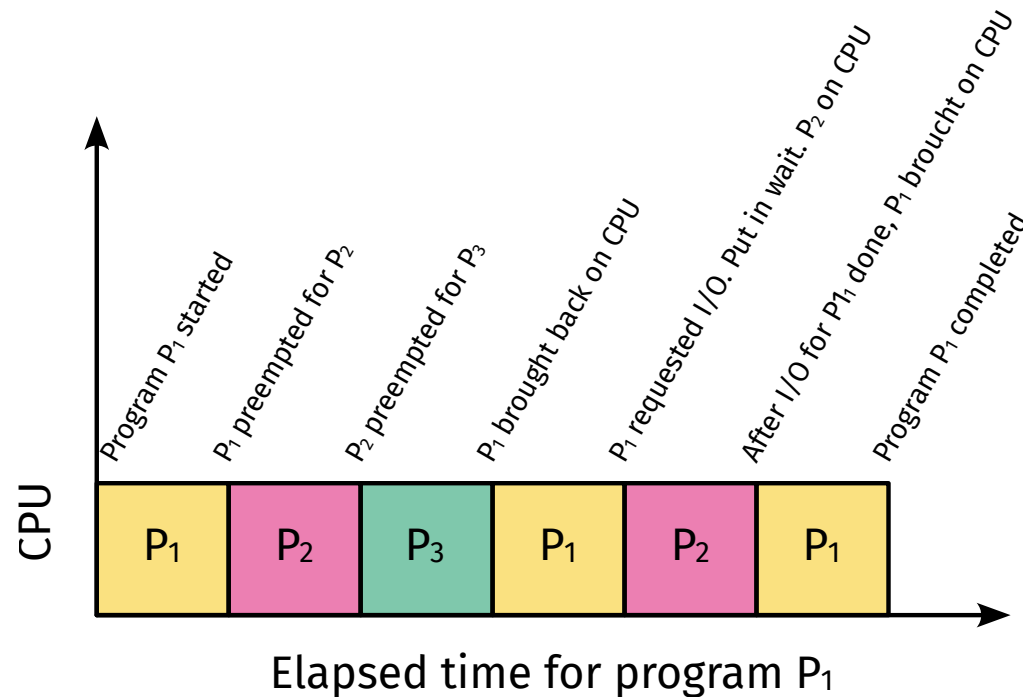
# Why concurrency?

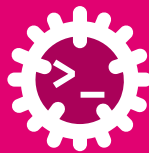


A program that does **one thing at a time** leaves the system underutilized.

- Programs spend a lot of time **waiting**
  - disk, network, the user
- That waiting time could be spent doing **other useful work**.

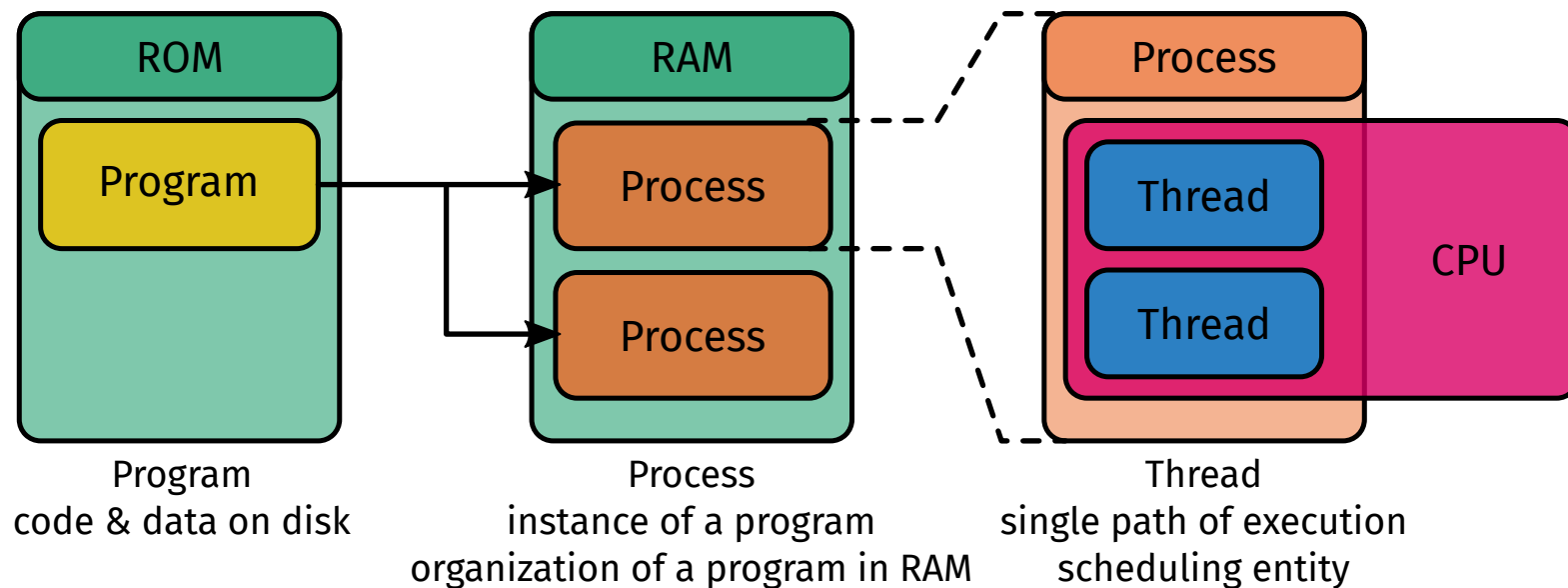
**Goal:** do more in the same wall-clock time, and stay **responsive**.





Three words we will use constantly - let's pin them down first.

- **Program** - code & data sitting **on disk** (not running)
- **Process** - a **running** instance of a program, with its **own memory**
- **Thread** - a **single path of execution** inside a process (the unit the **OS schedules**)

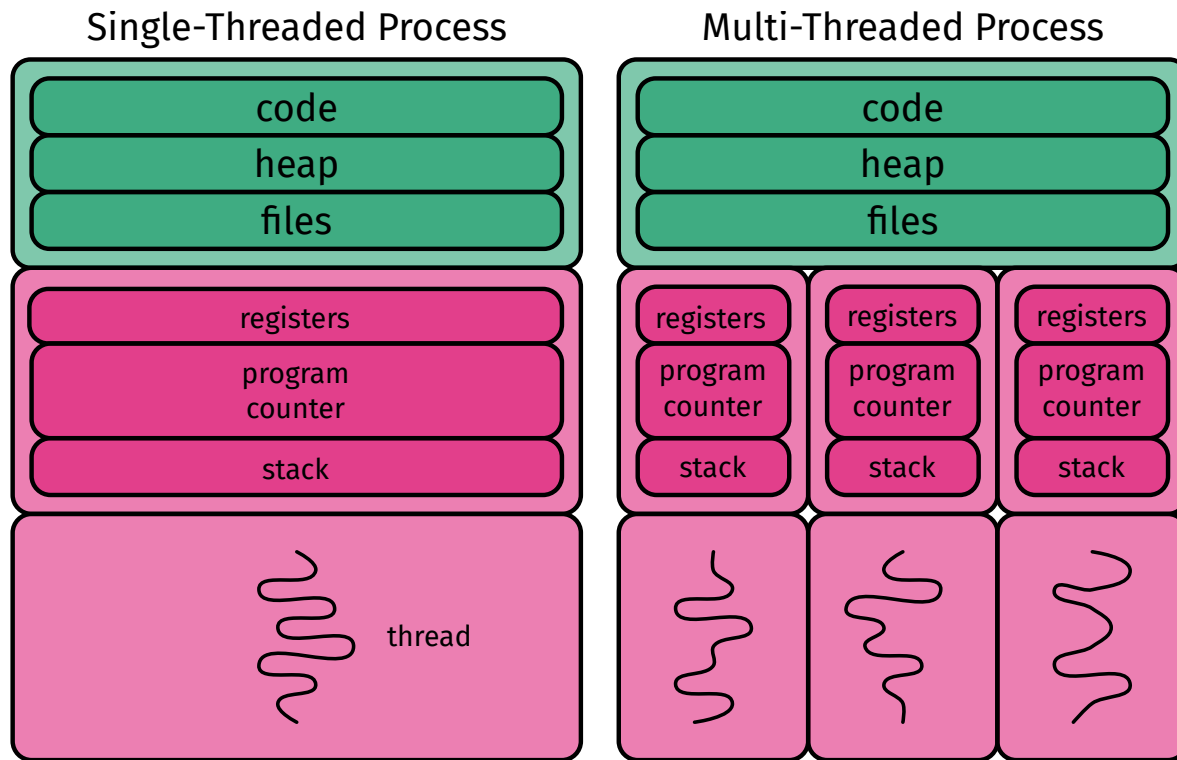


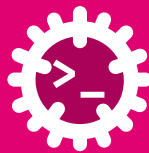




A process can hold **several threads**. They **share** most of the process...

- **Shared** by all threads
  - **code**, **heap**, open **files**
- **Private** to each thread
  - its own **stack** and **registers**
- Sharing the heap is what makes threads **fast to communicate**...
- ...and exactly what makes them **dangerous**





A **data race** occurs when:

1. two or more threads access the **same** memory location,
2. at least one of these operation is a **write** access,
3. the operations are **unsynchronized** (no lock, no ordering guarantee between them)
4. the operations **overlap in time**, meaning they occur concurrently.



The result is **undefined** - corrupted data, random crashes.

## Fearless concurrency

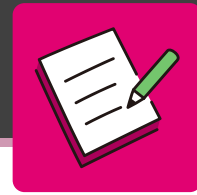
Rust rejects data races **at compile time**:

- **Ownership & borrowing**
  - one **&mut** **or** many **&**, never both
  - either **many readers**, or **one writer**, never both
- **Send** - a type is safe to **move** to another thread
- **Sync** - a type is safe to **share** (&T) between threads



If it **compiles**, it has **no data races**.

(Deadlocks & logic bugs 🐛 are still on you.)



Pick the **one** correct answer for each question.

### 1. What is private to each thread?

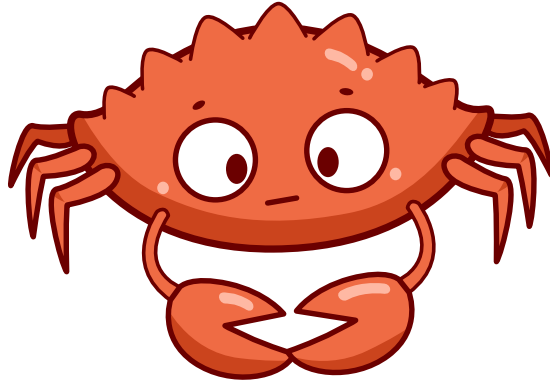
- a) the heap
- b) open files
- c) its stack and registers
- d) the program code

### 2. A data race needs all of these EXCEPT:

- a) two threads touch the same location
- b) at least one operation is a write
- c) the operations are guarded by a lock
- d) the operations overlap in time

### 3. “If it compiles, no data races” - thanks to:

- a) a garbage collector
- b) ownership / borrowing + Send / Sync
- c) the OS scheduler
- d) running on a single core

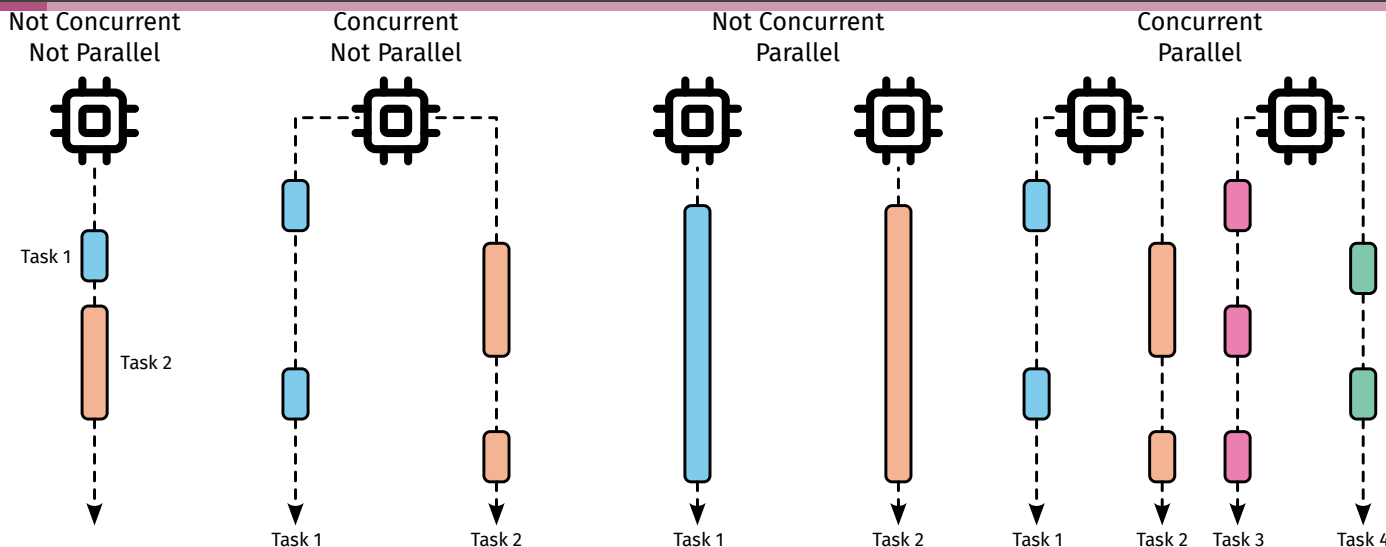
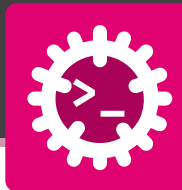


# Concurrency // Parallelism

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Structure vs. simultaneous execution

# Concurrency ≠ Parallelism



## Concurrency - structure

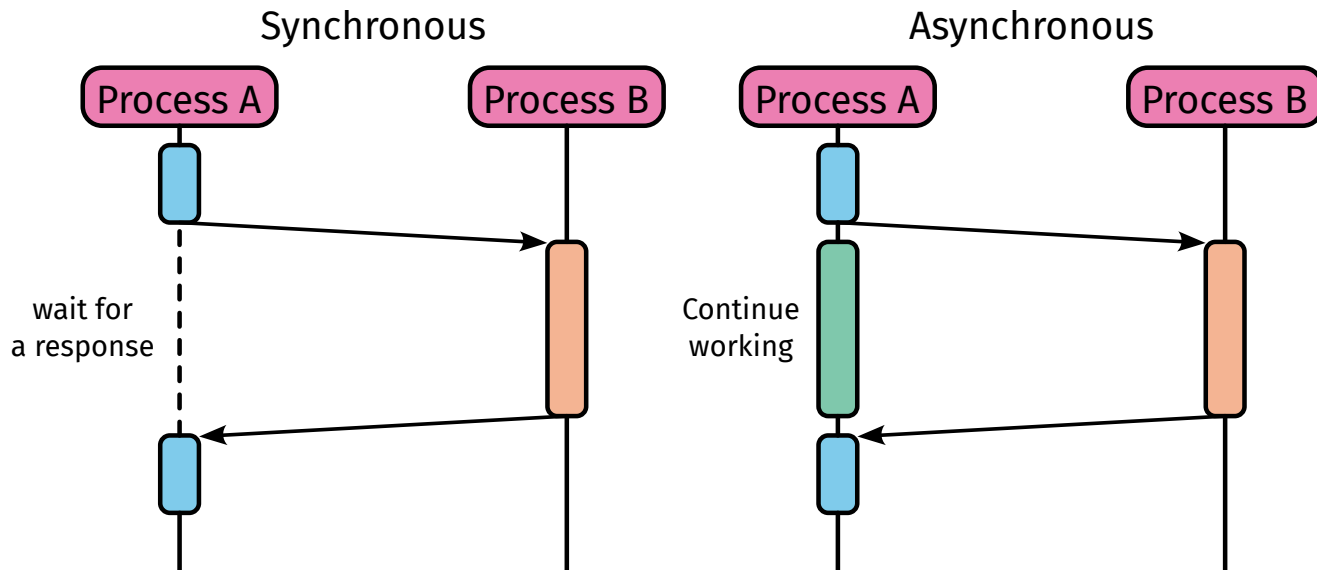
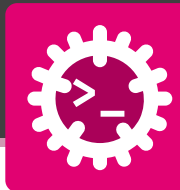
- many tasks that progress **independently**
- can **interleave** on a **single** core
- *“dealing with many things at once”*

## Parallelism - execution

- tasks running at the **same instant**
- needs **multiple** cores
- *“doing many things at once”*



A program can be **concurrent** on one core, and becomes **parallel** when given more.



## Synchronous

Process B must **wait** for Process A's response before continuing.

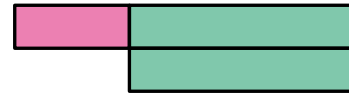
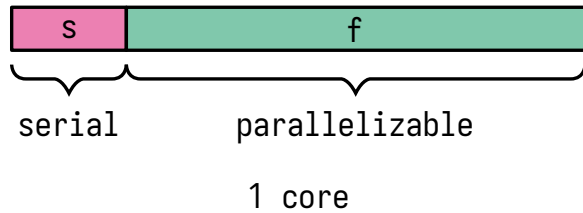
## Asynchronous

Process A **fires off** the request and **keeps working**, it handles the response later.

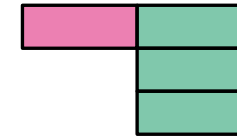


The same idea at every level: **don't block while you wait.**

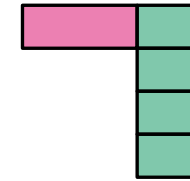
# The limit: Amdahl's Law



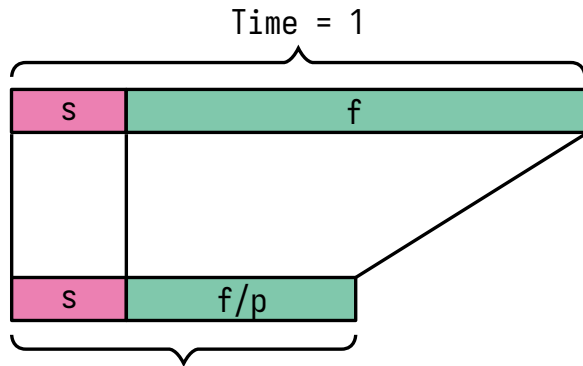
2 cores



3 cores



4 cores

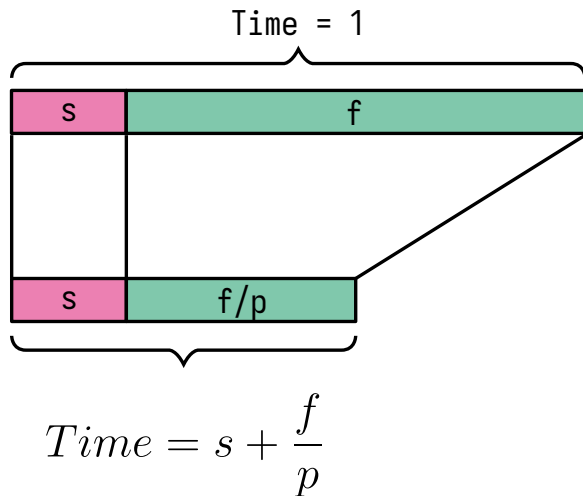
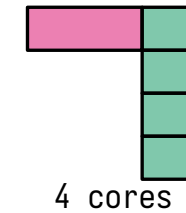
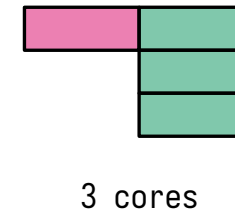
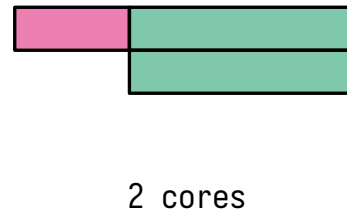
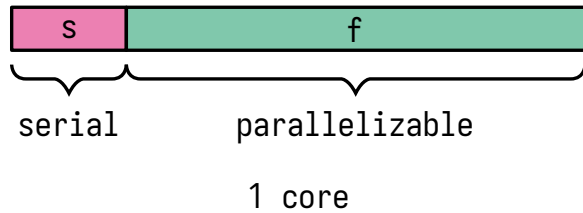


$$Time = s + \frac{f}{p}$$



If 50% of the execution time is sequential, the maximum speedup is 2x, no matter how many cores you throw at it.  
- Gene Amdahl, 1967

# The limit: Amdahl's Law



$$S = \frac{1}{s + \frac{f}{p}} = \frac{1}{1 - f + \frac{f}{p}}$$

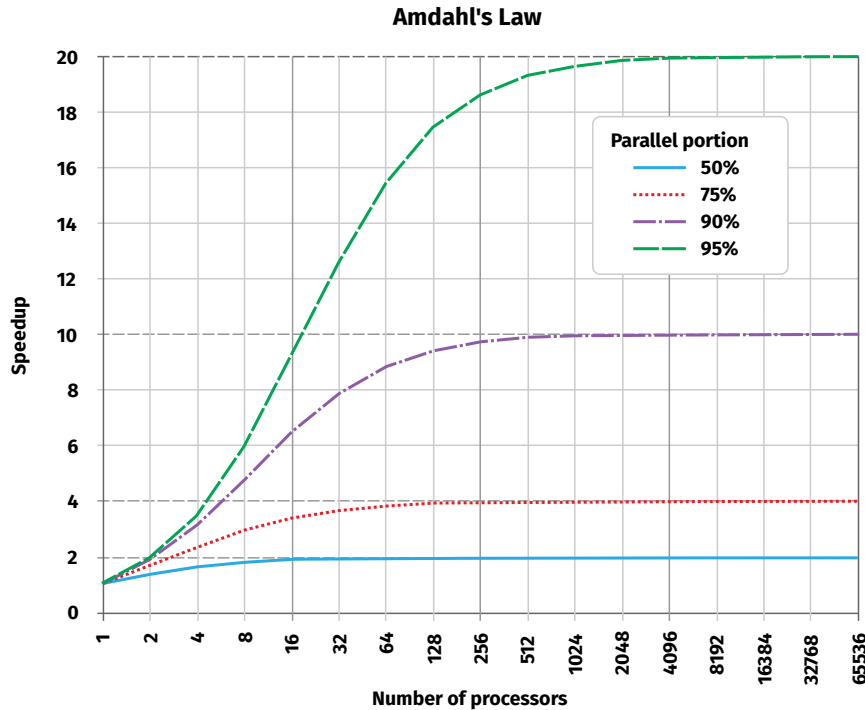
$S$  = Speedup

$s = 1 - f$  = Serial fraction

$f = 1 - s$  = Parallel fraction

$p$  = Number of processors

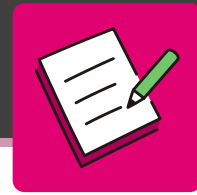




- Every program has a **serial part** that **cannot** be parallelised.
- Only the **parallel part** benefits from more cores.
- Example: if **5%** is serial, the maximum speed-up is **20×**, ever.



Adding cores has **diminishing returns**.  
Measure before you parallelise.



Floating point instructions improved to run 2x as fast, but only 10% of all executed instructions are Floating point. What will the overall speedup be?



# Threads

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Spawning and joining



```
use std::thread;
use std::time::Duration;

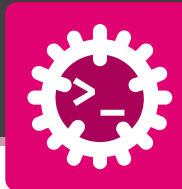
fn main() {
    // spawn a thread that runs ALONGSIDE main
    let handle = thread::spawn(|| {
        for i in 1..=4 {
            println!("  thread: step {i}");
            thread::sleep(Duration::from_millis(100));
        }
        42 // value returned by the thread
    });

    // main keeps working at the SAME time
    for i in 1..=4 {
        println!("main: step {i}");
        thread::sleep(Duration::from_millis(100));
    }

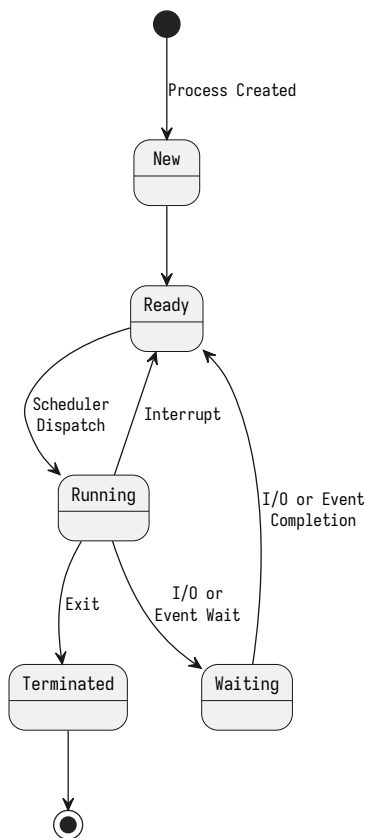
    // join: wait for the worker, collect its result
    let result = handle.join().unwrap();
    println!("thread returned {result}");
}
```

```
main: step 1
  thread: step 1
main: step 2
  thread: step 2
main: step 3
  thread: step 3
main: step 4
  thread: step 4
thread returned 42
```

- `thread::spawn` starts a **new OS thread** that runs **alongside** main
- Both loops print **interleaved** - proof they run at the **same time**
- `join()` **blocks** until the thread finishes and **returns its value** (42)
- Without `join`, main could **end first** and kill the thread mid-run



A thread moves through a small **state machine** - the OS decides who runs when.



## The states a thread is in:

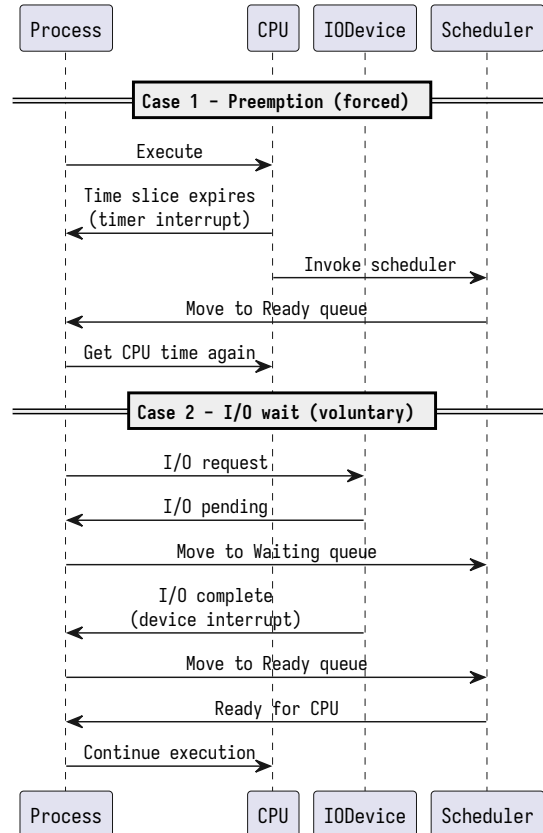
- **Ready** - requires the Central Processing Unit (CPU), waiting its turn
- **Running** - currently executing on a core
- **Waiting** - blocked on Input/Output (I/O) or an event, not competing for the CPU

## How it moves between them:

- **Dispatch** - the scheduler picks a Ready thread and runs it
- **Preemption** - the OS can **interrupt** a Running thread at any time, sending it back to Ready so another can run
- **Block / wake** - a thread that waits on I/O leaves the CPU when the I/O completes it returns to Ready



A running thread gives up the core in **two** ways - the scheduler then parks it and runs another.



## Case 1 - preemption (forced):

- The thread runs on the **CPU** until its **time slice** expires
- The **CPU invokes the scheduler**, which moves the thread to the **ready queue**
- It waits its turn, then gets the **CPU** back

## Case 2 - I/O wait (voluntary):

- The thread issues an **I/O request** and can't continue
- It moves to the **waiting queue** - off the **CPU**, not competing for it
- When the device signals **completion** (an **interrupt**), the thread returns to the **ready queue**



Either way the **CPU** is never idle - while one thread waits, another runs.



```
use rayon::prelude::*;
use std::time::Instant;

fn cost(n: u64) → u64 {
    (1..=n).map(|i| i % 7).sum()
}

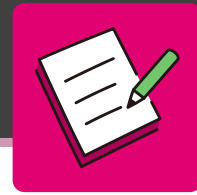
fn main() {
    let nums: Vec<u64> = (1..=20_000).collect();

    // — sequential —
    let start = Instant::now();
    let _seq: u64 = nums.iter().map(|&n| cost(n)).sum();
    let seq_time = start.elapsed();

    // — parallel (Rayon) —
    let start = Instant::now();
    let _par: u64 = nums.par_iter().map(|&n| cost(n)).sum();
    let par_time = start.elapsed();
    let speedup = seq_time.as_secs_f64() / par_time.as_secs_f64();
    println!("sequential: {seq_time:?}");
    println!("parallel:    {par_time:?}");
    println!("speedup:      {:.2}x", speedup);
}
```

```
# Macbook Pro M4 Max (16cores)
sequential: 1.553656209s
parallel:   133.673667ms
speedup:    11.62x
```

- Compare `iter()` (sequential) vs `par_iter()` (parallel), timed with `Instant`
  - One line changes - `iter()`  $\Rightarrow$  `par_iter()` - and Rayon splits the work over a **thread pool** ( $\approx$  one thread per core)
  - **11.6× faster** here on a CPU-bound, **independent** workload
  - Rust's safety guarantees still hold - **no data races**, enforced at compile time



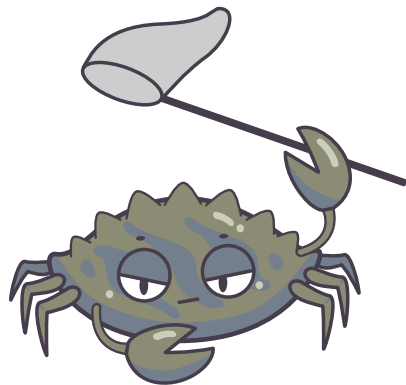
**Read** the program and give the value it prints. Does the **thread order** change the result?

```
use std::thread;

fn main() {
    let mut handles = vec![];
    for i in 1..=3 {
        handles.push(thread::spawn(move || i * 10));
    }

    let mut total = 0;
    for h in handles {
        total += h.join().unwrap();
    }
    println!("total = {total}"); // ?
}
```



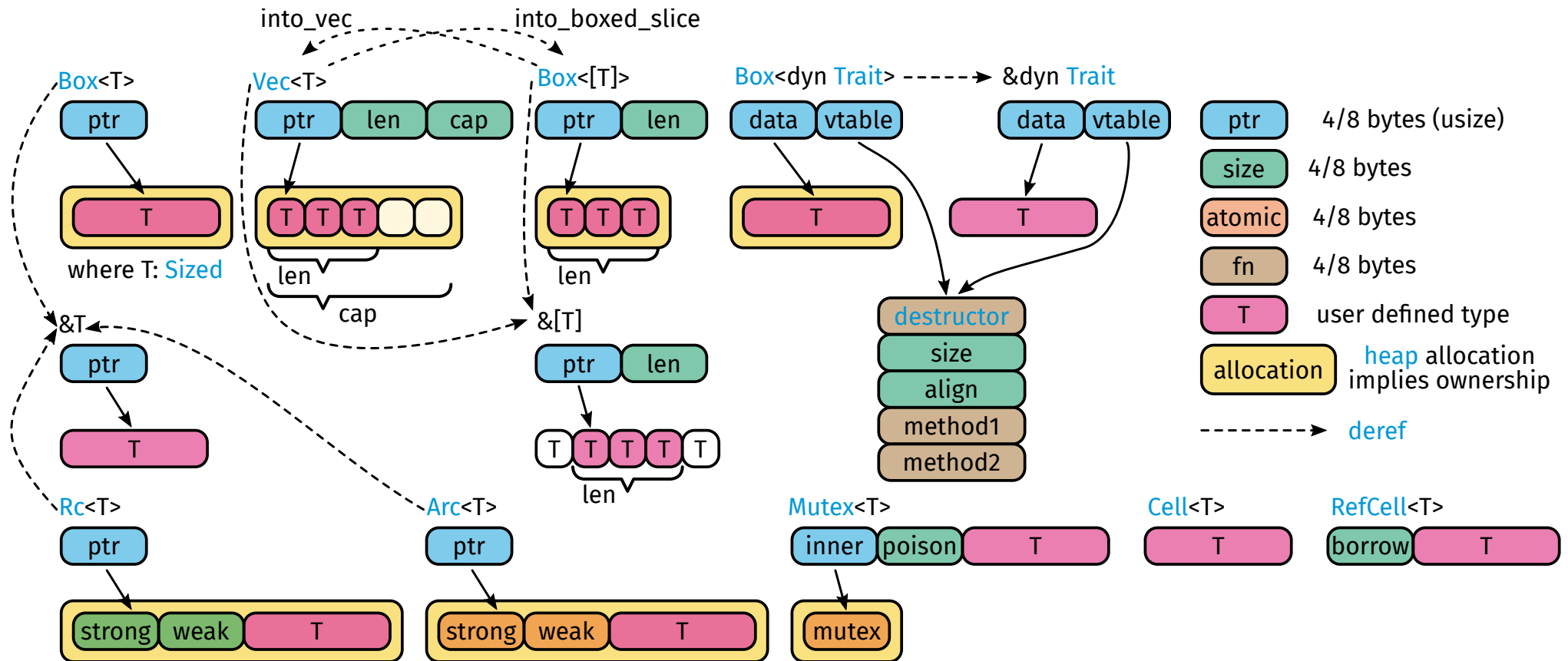


# Smart Pointers

---

Owning and sharing values safely

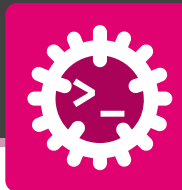
# Smart Pointers Overview



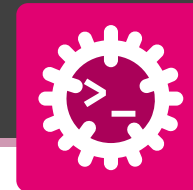


A **smart pointer** is a type that owns data **and adds behaviour** (heap allocation, reference counting, locking, runtime borrow checks). A plain `&T` is just a **borrow** - not a smart pointer.

| Type         | Group                             | Description                                                                                                                                                                                                                    |
|--------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Single owner | <code>Box&lt;T&gt;</code>         | Puts a value on the <b>heap</b> instead of the stack, with <b>one</b> owner. Use it for data too big for the stack, or for <b>recursive</b> types (a struct that contains itself) where the size would otherwise be infinite.  |
|              | <code>Box&lt;dyn Trait&gt;</code> | A Box holding a <b>trait object</b> : “some type implementing <code>Trait</code> , decided at run time.” Lets you own a value when you don’t know its concrete type - e.g. one element in a list of mixed shapes.              |
| Shared owner | <code>Rc&lt;T&gt;</code>          | <b>Reference-counted</b> shared ownership: many parts of your program co-own one value. It counts the owners and frees the value when the <b>last</b> one is dropped. <b>Single thread only</b> (the count isn’t thread-safe). |
|              | <code>Arc&lt;T&gt;</code>         | The same as <code>Rc</code> , but the owner count mutations are <b>atomic</b> , so it is safe to share <b>across threads</b> . Slightly slower than <code>Rc</code> ; use it only when threads are involved.                   |



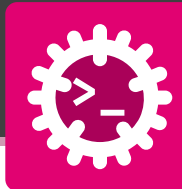
| Type                | Group      | Description                                                                                                                                                                                                                                                                         |
|---------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interior mutability | Cell<T>    | Lets you <b>change</b> a value even through a shared & reference, by swapping the whole value in and out with <code>get</code> / <code>set</code> . For small Copy types (numbers, flags). No borrowing, never panics.                                                              |
|                     | RefCell<T> | Like <code>Cell</code> , but lets you <b>borrow</b> the inner value ( <code>borrow</code> / <code>borrow_mut</code> ). The borrow rules (one writer or many readers) are checked <b>while the program runs</b> instead of at compile time - and it <b>panics</b> if you break them. |
|                     | Mutex<T>   | Guards data so <b>one thread at a time</b> can touch it. <code>lock()</code> blocks until it's your turn, hands you exclusive access, and <b>unlocks automatically</b> when you're done. The way to mutate shared data across threads.                                              |
|                     | RwLock<T>  | Like <code>Mutex</code> , but smarter about reading: it allows <b>many readers at once</b> OR <b>one writer</b> . Use it when reads are frequent and writes are rare.                                                                                                               |



Ask yourself two questions: **How many threads?** and **read or write?**

| Need                                                        | Single thread  | Multi-thread                 |
|-------------------------------------------------------------|----------------|------------------------------|
| heap, <b>one owner</b>                                      | Box<T>         | Box<T>                       |
| <b>shared</b> ownership (read)                              | Rc<T>          | Arc<T>                       |
| <b>shared + mutable</b>                                     | Rc<RefCell<T>> | Arc<Mutex<T>> Arc<RwLock<T>> |
| <b>interior mutability</b><br>mutate a Copy field via &self | Cell<T>        | Atomic*                      |

- The columns **mirror** each other: Rc ⇔ Arc, RefCell ⇔ Mutex, Cell ⇔ Atomic
- Same three ideas - **own, share, mutate-through-shared** - each with a **thread-safe** twin
- **Rule**: pick the **single-thread** tool unless you actually cross threads (it's cheaper)



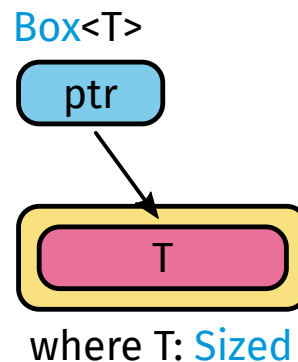
**Concept:** put **one** value on the **heap**.

- The **same** tool single- or multi-threaded (move it across with move).
- The value is freed when the Box drops (RAII).

```
use std::thread;
// 4 KB grid: too big for the stack arrow.stroked heap
let grid = Box::new([[0u8; 64]; 64]);
// `move` transfers the ONE owner to the worker
let h = thread::spawn(move || grid.len());
println!("rows: {}", h.join().unwrap()); // 64
```

## Box<T>

- one value on the **heap**  
(data too big for the stack, or a **recursive** type)
- move hands that **sole owner** to a thread





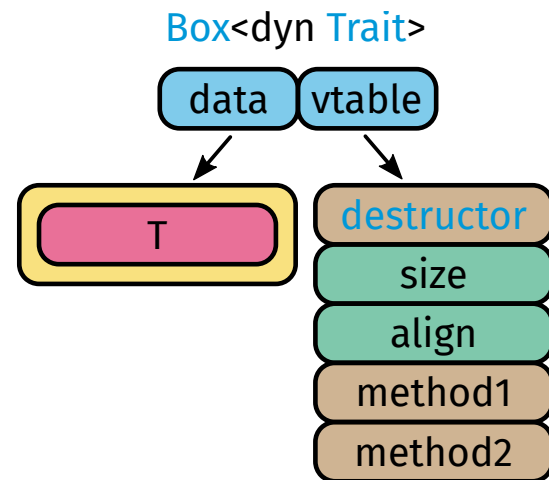
**Concept:** put **one** value on the **heap**.

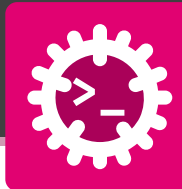
- The **same** tool single- or multi-threaded (move it across with move).
- The value is freed when the Box drops (RAII).

```
trait Job: Send { fn run(&self); }
struct Email;
impl Job for Email {
    fn run(&self) { println!("sending"); }
}
// concrete type hidden behind the Box
let job: Box<dyn Job> = Box::new(Email);
// move the trait object onto a worker thread
thread::spawn(move || job.run()).join().unwrap();
```

## Box<dyn Trait>

- owns **some** type implementing Trait, picked at **run time** (dynamic dispatch).
- A Send object can move across threads.





**Concept:** many owners of **one** read-only value

- a non-atomic **reference count** frees it when the **last** owner drops.

```
use std::rc::Rc;

struct Config { db_url: String, retries: u32 }
struct Service { cfg: Rc<Config> }

fn main() {
    // build the read-only config ONCE
    let cfg = Rc::new(Config {
        db_url: "postgres://db:5432".into(),
        retries: 3,
    });

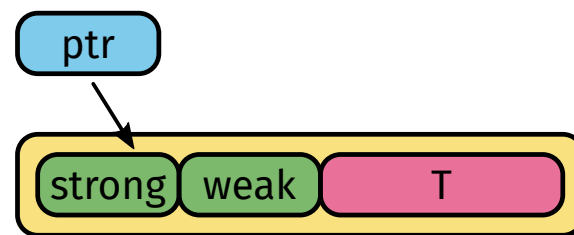
    // every service co-owns the SAME config, no data copy
    let api = Service { cfg: Rc::clone(&cfg) };
    let worker = Service { cfg: Rc::clone(&cfg) };

    println!("owners: {}", Rc::strong_count(&cfg)); // 3
    println!("{}", api.cfg.db_url, worker.cfg.retries);
}
```

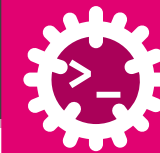
## Rc<T>

- many owners of one **immutable** value
- clone bumps a **non-atomic** count  
⇒ **no deep copy**
  - last owner dropped ⇒ value **freed**
- **single thread only** (count isn't thread-safe)

Rc<T>







**Concept:** **many** owners of **one** read-only value, but the count is **atomic**  $\Rightarrow$  safe to share **across threads**

- Slightly slower, use it only when threads are involved.

```
use std::sync::Arc;
use std::thread;

fn main() {
    // load a big read-only dataset ONCE
    let data = Arc::new(vec![7, 2, 9, 4, 8, 1, 5, 3]);
    let mut handles = vec![];
    for t in 0..4 {
        let data = Arc::clone(&data); // owner per thread
        handles.push(thread::spawn(move || {
            // each thread reads its own slice, no copy
            data[t * 2..t * 2 + 2].iter().sum::<i32>()
        }));
    }

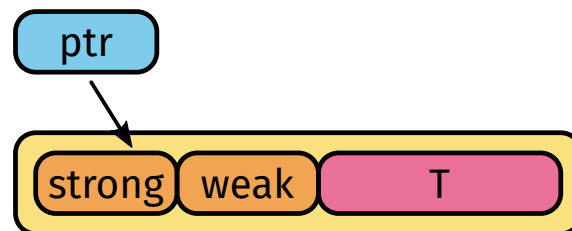
    let total: i32 = handles.into_iter()
        .map(|h| h.join().unwrap()).sum();
    println!("total = {total}"); // 39
}
```

## Arc<T>

- **atomic** reference count  $\Rightarrow$  Send + Sync
- share **immutable** data across threads, **no copy**
- each thread gets its own **owner** via **clone**
- need to **mutate**? add a lock

$\Rightarrow$  Arc<Mutex<T>>

Arc<T>





Every Rc/Arc carries **two** counts. The value lives as long as **strong > 0**.

```
use std::rc::{Rc, Weak};

let a = Rc::new(5);
let b = Rc::clone(&a);           // strong owner
let w: Weak<i32> = Rc::downgrade(&a); // weak, non-owning

println!("{}", Rc::strong_count(&a)); // 2
println!("{}", Rc::weak_count(&a));  // 1

// a Weak must be upgraded before use
if let Some(rc) = w.upgrade() {    // Some while alive
    println!("still alive: {rc}"); // 5
}                                  // None once strong = 0
```

- **Strong** (strong\_count) - the **owning** handles from new/clone. When it hits **0**, the value is **dropped**.
- **Weak** (weak\_count) - **non-owning** handles from downgrade. They **do not** keep the value alive.
- upgrade() turns a Weak into an Option<Rc> - None if the value is already gone, so you can never read freed memory.



Use Weak to break **reference cycles** (e.g. child  $\Rightarrow$  parent): a cycle of **strong** Rcs never reaches 0 and **leaks**. Arc works identically via Arc::downgrade.



## Concept:

- Rc gives **many owners** but only shared **reading**.
- Put a RefCell inside to also **mutate**  $\Rightarrow$  borrow rules then checked **at runtime**.

```
use std::rc::Rc;
use std::cell::RefCell;

fn main() {
    // one bank account, co-owned by two parts of the app
    let account = Rc::new(RefCell::new(0));
    let teller = Rc::clone(&account); // second owner

    *account.borrow_mut() += 100; // deposit
    *teller.borrow_mut() -= 30; // withdraw

    // both handles see the same balance
    println!("balance: {}", account.borrow()); // 70
}
```

## Rc<RefCell<T>>

- **single thread**: shared **and** mutable
- Rc - **many owners** of one value
- RefCell - mutate through &, rules checked **at runtime** (**panics** if you alias &mut)

RefCell<T>





## Concept: the **same shape across threads**

- Arc shares ownership atomically
- Mutex lets **one thread at a time** mutate the value safely.

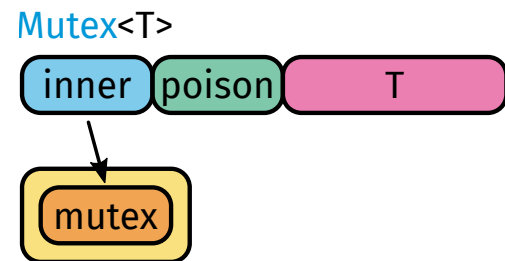
```
use std::sync::{Arc, Mutex};
use std::thread;

fn main() {
    // same account, now shared across worker threads
    let account = Arc::new(Mutex::new(0));

    let mut handles = vec![];
    for _ in 0..3 {
        let account = Arc::clone(&account); // owner per thread
        handles.push(thread::spawn(move || {
            *account.lock().unwrap() += 100; // safe deposit
        }));
    }
    for h in handles { h.join().unwrap(); }
    println!("balance: {}", *account.lock().unwrap()); // 300
}
```

## Arc<Mutex<T>>

- **across threads**: shared **and** mutable
- Arc - **atomic** many-owners (Send+Sync)
- Mutex - lock() waits its turn, grants **exclusive** access, guard **auto-unlocks**





**Concept:** mutate a value **through a shared &**

- no `&mut`, no `Rc`, no lock. The lightweight tool for a single `Copy` field (a counter, a flag).

```
use std::cell::Cell;
struct Button { clicks: Cell<u32> }
impl Button {
    // &self, yet we still mutate the counter
    fn click(&self) {
        self.clicks.set(self.clicks.get() + 1);
    }
    fn count(&self) -> u32 { self.clicks.get() }
}
fn main() {
    let b = Button { clicks: Cell::new(0) };
    b.click();
    b.click();
    println!("clicks: {}", b.count()); // 2
}
```

## `Cell<T>`

- mutate through `&self` (no `&mut`)
- get/set **swap the whole value**
- for small **Copy** types, **never panics**
- **single thread only**

`Cell<T>`

T



A `Cell` holds one **Copy** value.

To mutate a **non-Copy** value (a `Vec`, a `String`) through `&`, reach for `RefCell`.



**Concept:** the **lock-free** twin of Cell for one Copy value

- a single hardware instruction does the update, safe to share across threads via Arc.

```
use std::sync::Arc;
use std::sync::atomic::{AtomicU32, Ordering};
use std::thread;

fn main() {
    // shared hit counter, no lock needed
    let hits = Arc::new(AtomicU32::new(0));

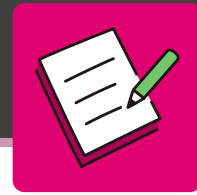
    let mut handles = vec![];
    for _ in 0..4 {
        let hits = Arc::clone(&hits);
        handles.push(thread::spawn(move || {
            for _ in 0..25 {
                hits.fetch_add(1, Ordering::Relaxed);
            }
        }));
    }
    for h in handles { h.join().unwrap(); }
    println!("hits: {}", hits.load(Ordering::Relaxed)); // 100
}
```

## Atomic\*

- AtomicU32, AtomicBool, AtomicUsize, ...
- **lock-free**: one atomic CPU instruction
- share across threads with Arc
- perfect for **counters** and **flags**



Atomics fit only **Copy** values. For a **non-Copy** shared value use Mutex/ RwLock. (Pin<P> and Cow<'a, T> exist too, but are out of scope here.)

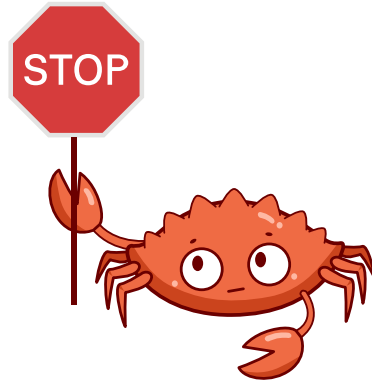


For each situation, name the **smart pointer** you would reach for.

1. A **single owner** of a large value that must live on the **heap**.
2. **Several owners** reading the same data, all on **one** thread.
3. **Several threads** that must **share and mutate** one value.
4. **Shared and mutable** data among many owners, on **one** thread.



Two questions decide it: **how many threads?** and **read or write?**

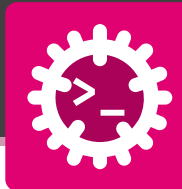


# Synchronization & Hazards

---

Locks, races and deadlocks





**Critical section** - code that must run **exclusively**, one thread inside at a time.

- Guard it with a `Mutex<T>` (or `RwLock<T>` for read-mostly data).
- Keep it **short** - it **serialises** your program (remember Amdahl!).
- Limit to **N** concurrent users? Use a **semaphore** (`tokio::sync::Semaphore`).

**Deadlock** - two threads wait **forever** on each other's locks:

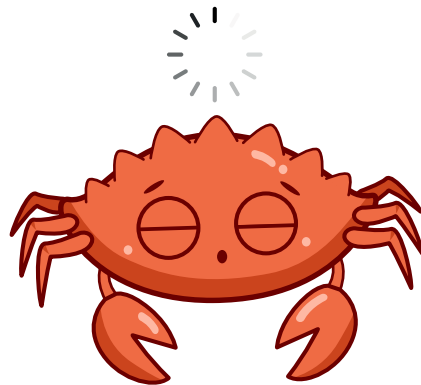
- Thread A holds lock 1, wants lock 2
- Thread B holds lock 2, wants lock 1
- Neither can proceed  $\Rightarrow$  **frozen**.

**How to avoid it**

- **Lock ordering** - always acquire locks in the **same order**
- **Timeouts** - `try_lock` and back off
- **Less shared state** - prefer **message passing**



Rust stops **data races**, **not** deadlocks  
- those are a **design** problem.



# Asynchronous Execution Model

---

Make progress while you wait



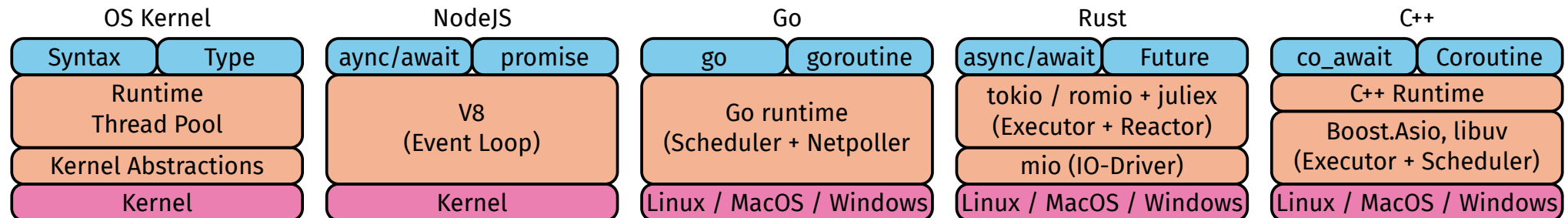
**Asynchronous execution is not threads** - it is a **separate model**: many **tasks** driven by a **runtime**, instead of **threads** scheduled by the **OS**.

**Threads** - the **OS** way

- one **stack** per task (~MB), **preemptively** scheduled
- **tens of thousands** of connections  $\Rightarrow$  as many threads
- most sit **blocked on I/O**, burning memory

**Async** - the runtime way

- thousands of **tasks** on a **handful** of threads
- a task is a cheap **state machine**, **cooperatively** scheduled
- ideal for **I/O-bound** waiting at scale





Write code that **reads** sequentially but **doesn't block** the thread.

```
// `async fn` returns a Future instead of running now
async fn fetch(id: u32) → String {
    // .await yields the thread while the IO is pending
    let body = download(id).await;
    format!("got {} bytes", body.len())
}

#[tokio::main]
async fn main() {
    // both tasks share ONE thread, interleaving at .await
    let (a, b) = tokio::join!(fetch(1), fetch(2));
    println!("{a} / {b}");
}
```

- `async fn` - turns code into a **Future**
- `.await` - “**pause here** until ready, **let others run** meanwhile”
- `#[tokio::main]` - starts the **runtime** that drives it all



**Idea:** run **thousands** of tasks on a **handful** of threads - each one **yields** at `.await` instead of blocking.

```
async fn handler() {  
    // ✗ blocks the WHOLE thread - every other  
    //   task on it is frozen until this returns  
    std::thread::sleep(Duration::from_secs(1));  
  
    // ✔ yields the thread, others make progress  
    tokio::time::sleep(Duration::from_secs(1)).await;  
}
```

- **Preemptive** (OS threads) - the kernel can **interrupt** anytime.
- **Cooperative** (async) - a task runs until it **yields** (`.await`).
- **Never block** inside async: no `thread::sleep`, no heavy CPU loops.
- Offload CPU work to **threads** / **Rayon**, or use `spawn_blocking`.



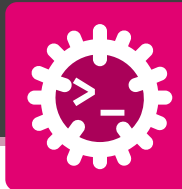
The compiler **rewrites** every `async fn` into a **state machine** - one state per `.await`.

```
async fn fetch(id: u32) → String {
    let body = download(id).await; // suspension point
    format!("got {} bytes", body.len())
}

// the compiler turns `fetch` into a state machine:
enum Fetch {
    Start { id: u32 },           // before the .await
    Downloading { fut: Download }, // parked at the .await
    Done,                       // value ready
}

#[tokio::main]
async fn main() {
    // the runtime polls that machine to completion
    let s = fetch(1).await;
    println!("{s}");
}
```

- Each `.await` is a **suspension point**  $\Rightarrow$  a **state**
- `poll` advances the machine **one state at a time**
- Locals alive **across** an `.await` are **stored in the future**
- Nothing is blocked - the task just **remembers where it was**



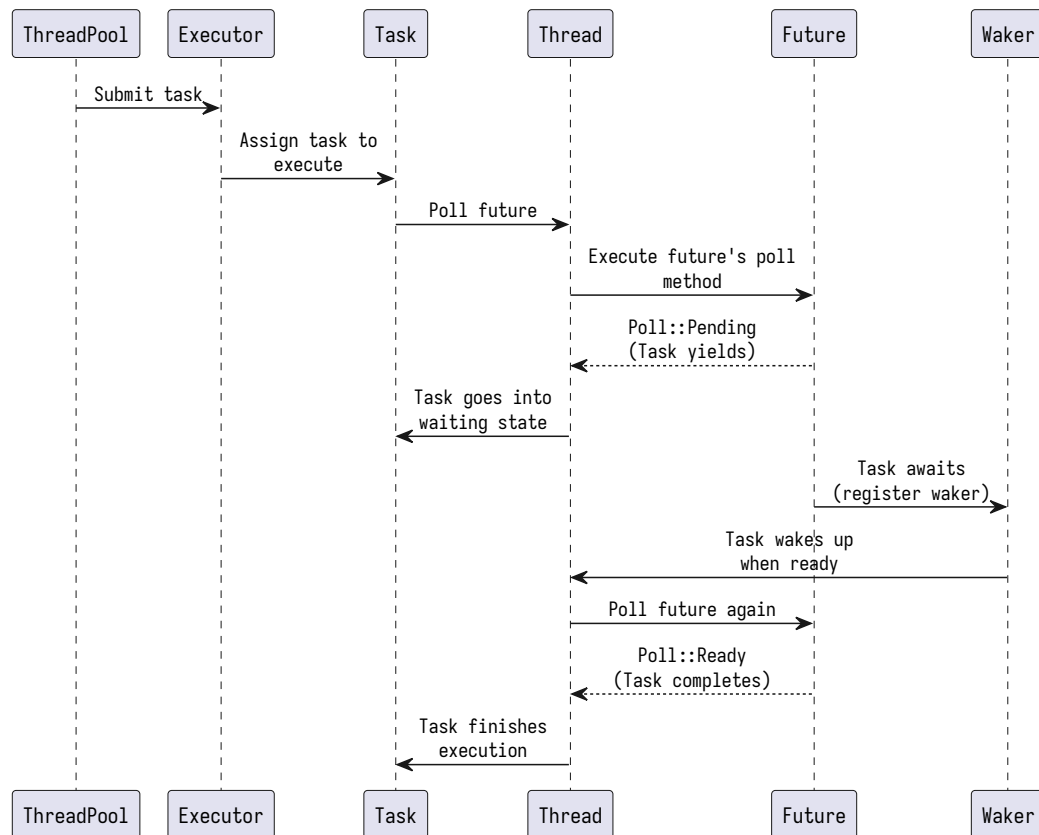
An async block **does nothing** until something **polls** it.

```
enum Poll<T> {  
    Ready(T),    // done, here is the value  
    Pending,     // not yet - poll me again later  
}  
  
// (simplified) a Future is polled to completion  
trait Future {  
    type Output;  
    fn poll(&mut self, cx: &mut Context)  
        → Poll<Self::Output>;  
}
```

- A Future is a **state machine** that is **polled**.
- Pending  $\Rightarrow$  not ready; Ready(v)  $\Rightarrow$  finished.
- **Lazy**: nothing runs until an **executor** polls it.
- So who polls, and how often?  $\Rightarrow$  the **runtime**.



The runtime is a handful of pieces that **poll your futures to completion**.



## Elements of a runtime

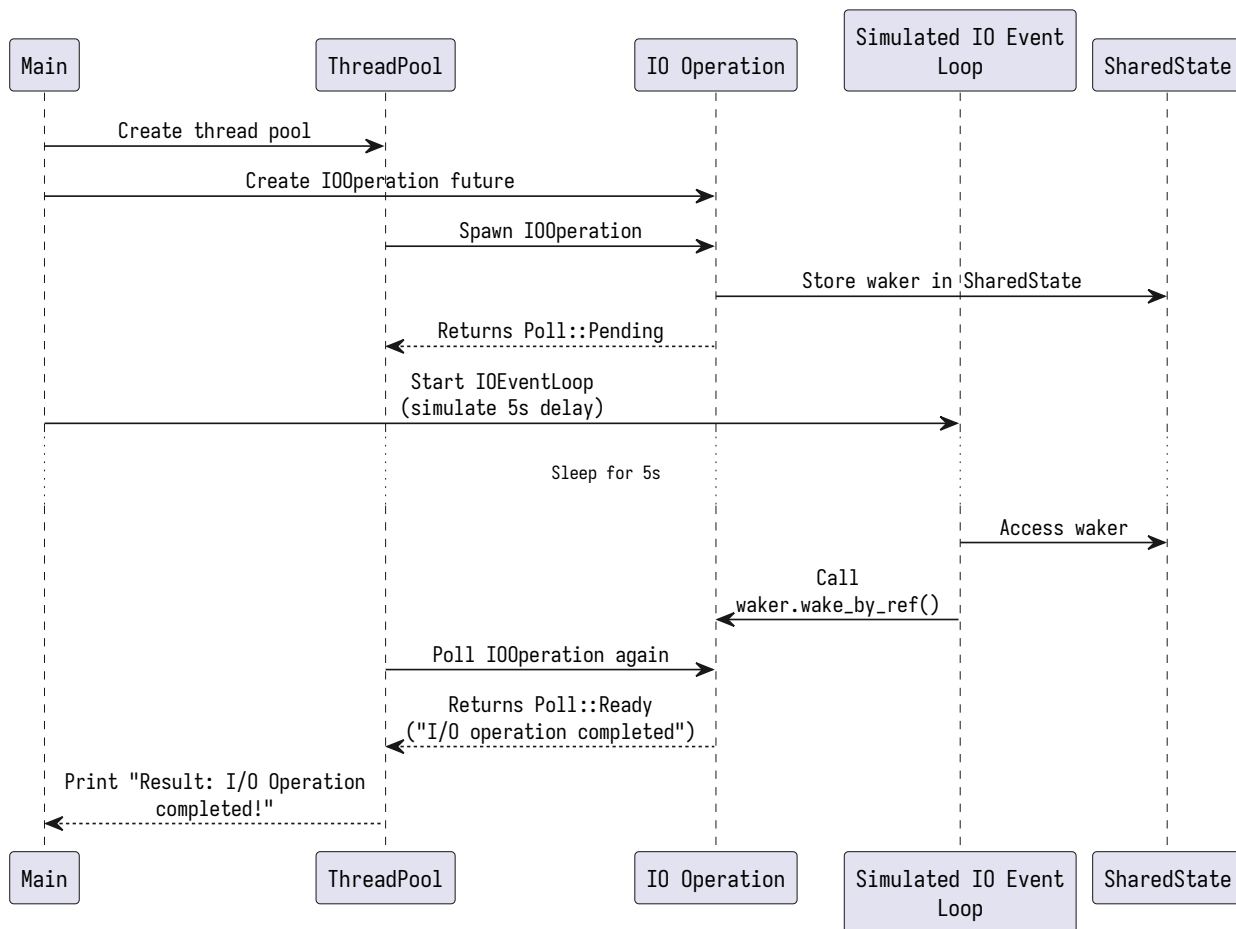
- **Executor** - polls futures on a **thread pool**
- **Reactor** - watches the **OS** for **I/O** readiness
- **Waker** - the “**call me back**” handle of a parked task

## The poll cycle

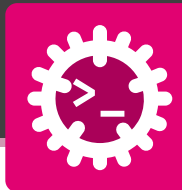
1. poll  $\Rightarrow$  Pending, **register a Waker**
2. task **parked** - uses **no CPU**
3. **I/O** ready  $\Rightarrow$  the **Waker** wakes it
4. poll again  $\Rightarrow$  Ready  $\Rightarrow$  done



# Under the hood: putting it together



- A parked task **stores its waker**
- when the **I/O** event fires, **wake()** schedules a **re-poll**  $\Rightarrow$  Ready.

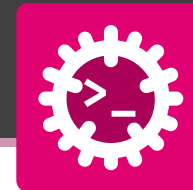
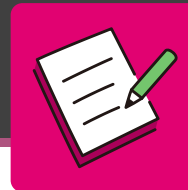


| Threads / Rayon                      | Async / Tokio                          |
|--------------------------------------|----------------------------------------|
| best for <b>CPU-bound</b> work       | best for <b>I/O-bound</b> work         |
| true <b>parallelism</b> across cores | huge <b>concurrency</b> on few threads |
| heavy: MB stack per thread           | cheap: a task is just a state machine  |
| spawn + join, par_iter               | async/.await, a <b>runtime</b>         |



## Rule of thumb

- compute  $\Rightarrow$  threads
- wait (network, disk)  $\Rightarrow$  async.



The function should **not block** the thread while it waits. Add the **two** missing pieces and answer the question.

```
// (1) make this function asynchronous
fn fetch(id: u32) → String {
  // (2) wait for the future without blocking the thread
  let body = download(id);
  format!("got {} bytes", body.len())
}
```



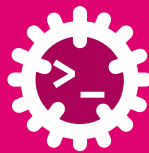
**Question:** why use `.await` here instead of a blocking call?



# Inter-Thread Communication

---

Talk, don't share



Threads need to **coordinate** and **hand data** to each other. There are **two ways**.

**Share memory** - with Smart Pointers, Mutexes, and RwLocks

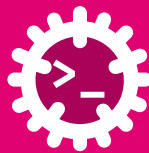
- one value behind `Arc<Mutex<T>>`
- every thread **locks**, touches it, unlocks
- risk of **races** (without the lock) and **deadlocks**

**Pass messages** - with channels

- threads **send owned values** down a **channel**
- the receiver **owns** what it gets - no shared lock
- sidesteps data races **and** most deadlocks



*“Don’t communicate by sharing memory; share memory by communicating.”* - the Go proverb, equally true in Rust.



From **threads in one process** to **separate processes** - same idea, different reach.

## Channels - between threads, one process

| Mechanism | What it is                                                                         |
|-----------|------------------------------------------------------------------------------------|
| mpsc      | <b>multi-producer, single-consumer</b> queue - many senders, one receiver (in std) |
| oneshot   | a <b>single</b> value sent <b>once</b> - perfect for a reply / result (Tokio)      |
| broadcast | <b>many consumers</b> each receive <b>every</b> message - fan-out (Tokio)          |

## OS-level Inter-Process Communication (IPC) - between processes

| Mechanism          | What it is                                                   |
|--------------------|--------------------------------------------------------------|
| pipes              | a <b>byte stream</b> from one process into another           |
| sockets            | a two-way channel, even <b>across machines</b> (the network) |
| files / shared mem | a <b>region</b> both processes read and write                |



The **most common** channel: many threads **send**, one thread **receives**.

```
use std::sync::mpsc;
use std::thread;

fn main() {
    let (tx, rx) = mpsc::channel(); // sender, receiver
    thread::spawn(move || {
        tx.send("hello from worker").unwrap(); // value MOVES in
    });
    let msg = rx.recv().unwrap(); // blocks until a message
    println!("{}", msg); // hello from worker
}
```

- `channel()` hands back a **sender** tx and **receiver** rx
- `send` **moves** the value into the channel - the sender gives it up
- `recv` **blocks** until a message arrives (or every sender is dropped)
- **clone tx** to get **multiple producers**



Channels turn **shared-state** problems into simple **producer / consumer** flows.



Let **3 worker threads** report their result to `main` through an `mpsc` channel - **no shared lock**.

1. Create a channel with `mpsc::channel()`.
2. Spawn **3** threads; each **clones** the sender and sends its `id * 10`.
3. In `main`, **receive** the 3 values and print their **sum**.



**Clone** tx for multiple producers; send **moves** the value into the channel; recv **blocks** until a message arrives.



WHY DO WHALES JUMP  
WHY ARE THERE MIRRORS ABOVE BEDS  
WHY DO I SAY UH  
WHY IS SEA SALT BETTER  
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS  
WHY IS THERE NOT A POKEMON MMO  
WHY IS THERE LAUGHING IN TV SHOWS  
WHY ARE THERE DOORS ON THE FREEWAY  
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING  
WHY AREN'T ANY COUNTRIES IN ANTARCTICA  
WHY ARE THERE SCARY SOUNDS IN MINECRAFT  
WHY IS THERE KICKING IN MY STOMACH  
WHY ARE THERE TWO SLASHES AFTER HTTP  
WHY ARE THERE CELEBRITIES  
WHY DO SNAKES EXIST  
WHY DO OYSTERS HAVE PEARLS  
WHY ARE DUCKS CALLED DUCKS  
WHY DO THEY CALL IT THE CLAP  
WHY ARE KYLE AND CARTMAN FRIENDS  
WHY IS THERE AN ARROW ON AANG'S HEAD  
WHY ARE TEXT MESSAGES BLUE  
WHY ARE THERE MUSTACHES ON CLOTHES  
WHY WUBA LUBBA DUB DUB MEANING  
WHY IS THERE A WHALE AND A POT FALLING  
WHY ARE THERE SO MANY BIRDS IN SWISS  
WHY IS THERE SO LITTLE RAIN IN WALLIS  
WHY IS WALLIS WEATHER FORECAST ALWAYS WRONG

WHY ARE THERE MALE AND FEMALE BIKES  
WHY ARE THERE BRIDESMAIDS  
WHY DO DYING PEOPLE REACH UP  
HOW FAST IS LIGHTSPEED  
WHY ARE OLD KLINGONS DIFFERENT

WHY ARE THERE  
SQUIRRELS



WHY AREN'T THERE DINOSAUR GHOSTS  
WHY DO IGUANAS DIE  
WHY ARE HATS SO EXPENSIVE  
WHY IS THERE CAFFEINE IN MY SHAMPOO  
WHY HAVE DINOSAURS NO FUR  
WHY AREN'T ECONOMISTS RICH  
WHY DO AMERICANS CALL IT SOCCER  
WHY ARE MY EARS RINGING  
WHY IS 42 THE ANSWER TO EVERYTHING  
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER  
WHY IS MARVIN ALWAYS SO SAD

WHY AREN'T ECONOMISTS RICH  
WHY DO AMERICANS CALL IT SOCCER  
WHY ARE MY EARS RINGING  
WHY IS 42 THE ANSWER TO EVERYTHING  
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER  
WHY IS MARVIN ALWAYS SO SAD

WHY IS EARTH TILTED  
WHY IS SPACE BLACK  
WHY IS OUTER SPACE SO COLD  
WHY ARE THERE PYRAMIDS ON THE MOON  
WHY IS NASA SHUTTING DOWN

WHY ARE THERE TINY SPIDERS IN MY HOUSE  
WHY DO SPIDERS COME INSIDE  
WHY ARE THERE HUGE SPIDERS IN MY HOUSE  
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE  
WHY ARE THERE SPIDERS IN MY ROOM  
WHY ARE THERE SO MANY SPIDERS IN MY ROOM  
WHY DO SPYDER BITES ITCH

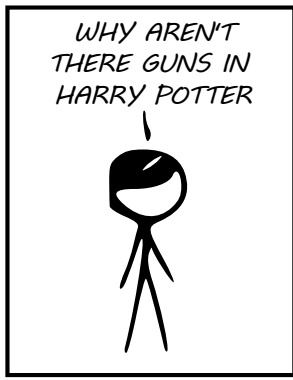
WHY ARE HATS SO EXPENSIVE  
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WHY ARE MY EARS RINGING  
WHY IS 42 THE ANSWER TO EVERYTHING  
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER  
WHY IS MARVIN ALWAYS SO SAD

# QUESTIONS

CAN BE ASKED BY ANYONE ANYTIME

WHY AREN'T ECONOMISTS RICH  
WHY DO AMERICANS CALL IT SOCCER  
WHY ARE MY EARS RINGING  
WHY IS 42 THE ANSWER TO EVERYTHING  
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER  
WHY IS MARVIN ALWAYS SO SAD

WHY ARE THERE FEMALE



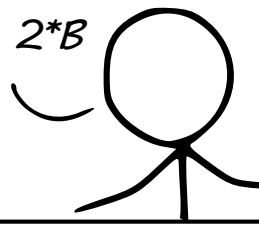
WHY AREN'T ECONOMISTS RICH  
WHY DO AMERICANS CALL IT SOCCER  
WHY ARE MY EARS RINGING  
WHY IS 42 THE ANSWER TO EVERYTHING  
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER  
WHY IS MARVIN ALWAYS SO SAD

WHY IS THERE AN OWL IN MY BACKYARD  
WHY IS THERE AN OWL OUTSIDE MY WINDOW  
WHY IS THERE AN OWL ON THE DOLLAR BILL  
WHY DO OWLS ATTACK PEOPLE  
WHY ARE FPGA's EVERYWHERE  
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE  
WHY ARE THERE GODS  
WHY ARE THERE TWO SPOCKS

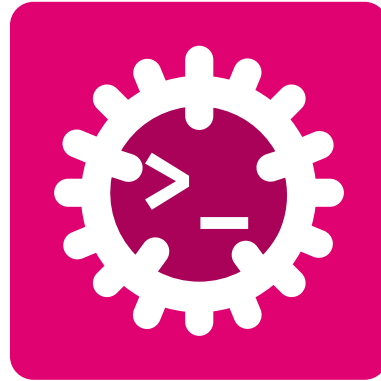
WHAT IS <https://xkcd.com/1256/>  
WHY DO THEY SAY T-MINUS  
WHY ARE OCEANS BECOMMING MORE ACIDIC

WHY IS HTTPS CROSSED OUT  
WHY IS THERE A LINE THROUGH  
WHY IS THERE A RED LINE THROUGH HTTPS  
WHY IS HTTPS IMPOR  
WHY IS YKK ON ALL ZIPPERS

WHY 2\*B / 2\*B



WHY ARE THERE SO MANY CROWS IN ROCHESTER  
WHY IS TO BE OR NOT TO BE FL  
WHY DO CHILDREN GET CANCER  
WHY IS POSEIDON ANGRY WITH ODYSSEUS  
WHY IS THERE ICE IN SPACE  
WHY IS THERE AN OWL IN MY BACKYARD  
WHY IS THERE AN OWL OUTSIDE MY WINDOW  
WHY IS THERE AN OWL ON THE DOLLAR BILL  
WHY DO OWLS ATTACK PEOPLE  
WHY ARE FPGA's EVERYWHERE  
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE  
WHY ARE THERE GODS  
WHY ARE THERE TWO SPOCKS  
WHY ARE MY BOOBS ITCHY  
WHY ARE CIGARETTES LEGAL  
WHY ARE THERE DUCKS IN MY POOL  
WHY IS JESUS WHITE  
WHY IS THERE LIQUID IN MY EAR  
WHY DO Q TIPS FEEL GOOD  
WHY DO PEOPLE DIE



## Glossary & Bibliography

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## Computer Science

**CPU – Central Processing Unit:** The main processing component of a computer. [20](#), [20](#), [20](#), [21](#), [21](#), [21](#), [21](#), [44](#), [44](#), [47](#), [49](#)

**OS – Operating System:** System software that manages hardware resources and applications. [1](#), [7](#), [10](#), [19](#), [20](#), [20](#), [20](#), [42](#), [42](#), [44](#), [47](#), [53](#)

## Concurrency

**I/O – Input/Output:** Operations that involve reading from or writing to external devices or resources. [20](#), [20](#), [20](#), [21](#), [21](#), [42](#), [42](#), [47](#), [47](#), [48](#), [49](#)

**IPC – Inter-Process Communication:** Mechanisms allowing processes to exchange data and synchronize execution. [53](#)



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